

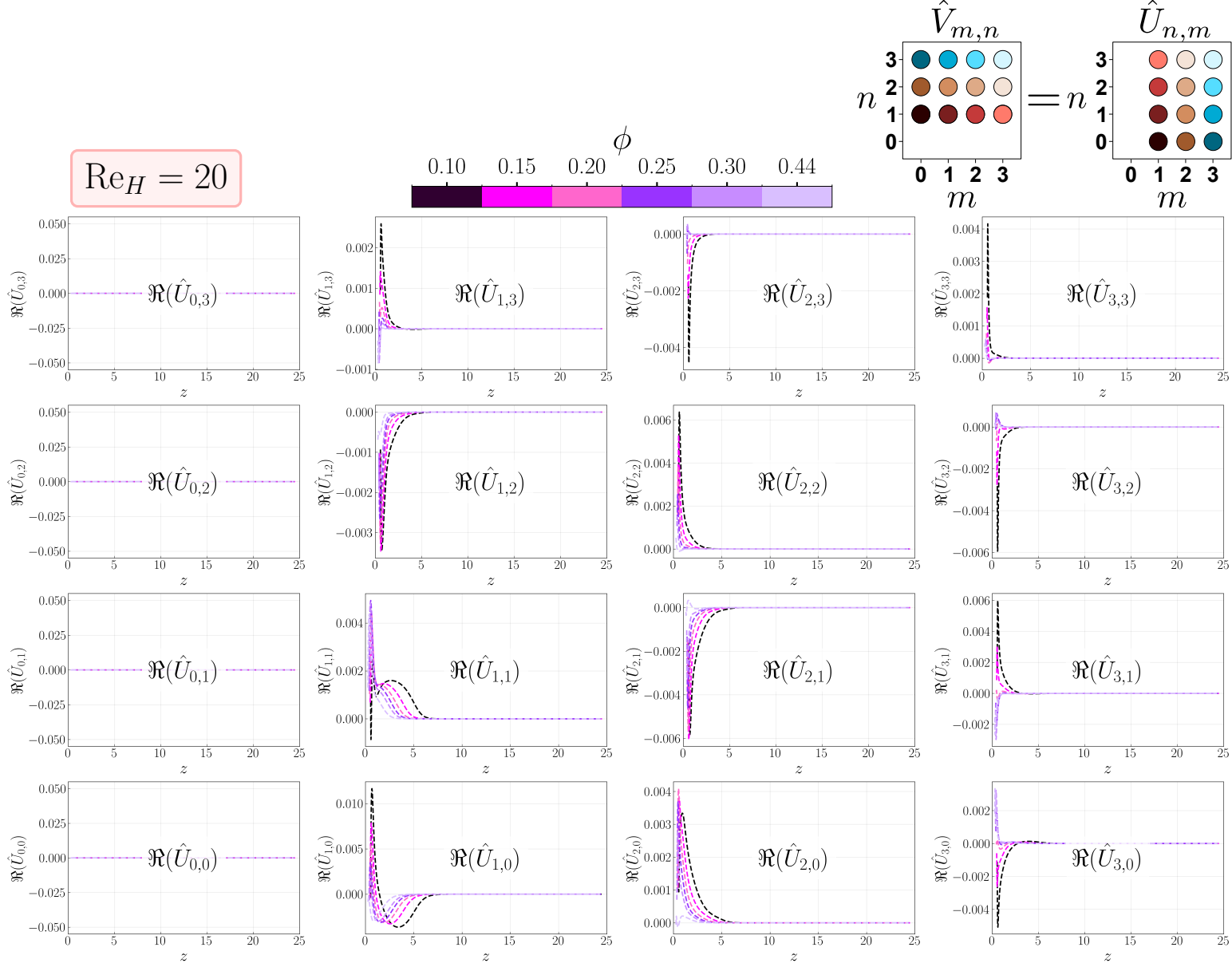


Figure B.53: Streamwise evolution of the real part $\Re(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 20$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

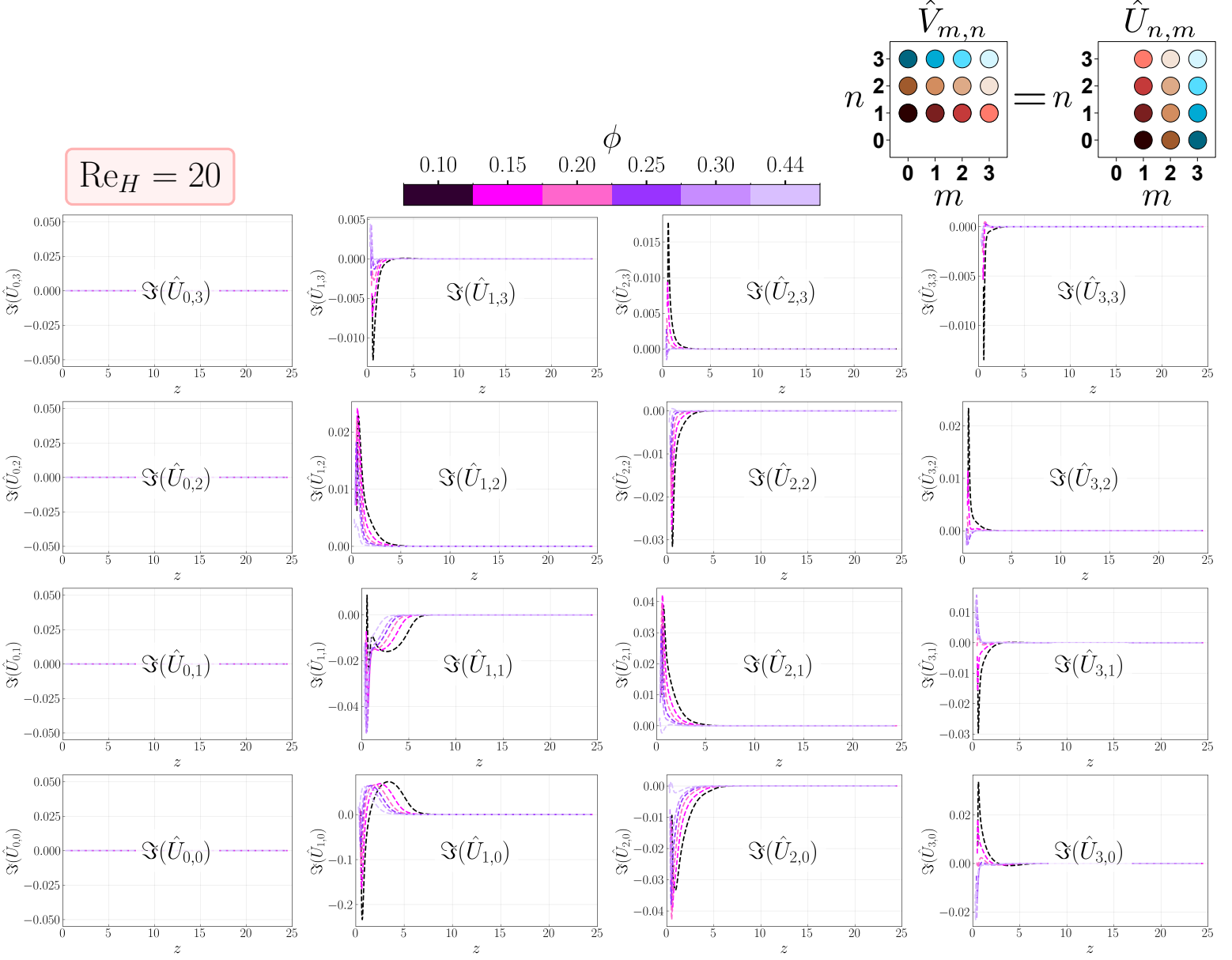


Figure B.54: Streamwise evolution of the imaginary part $\Im(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 20$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

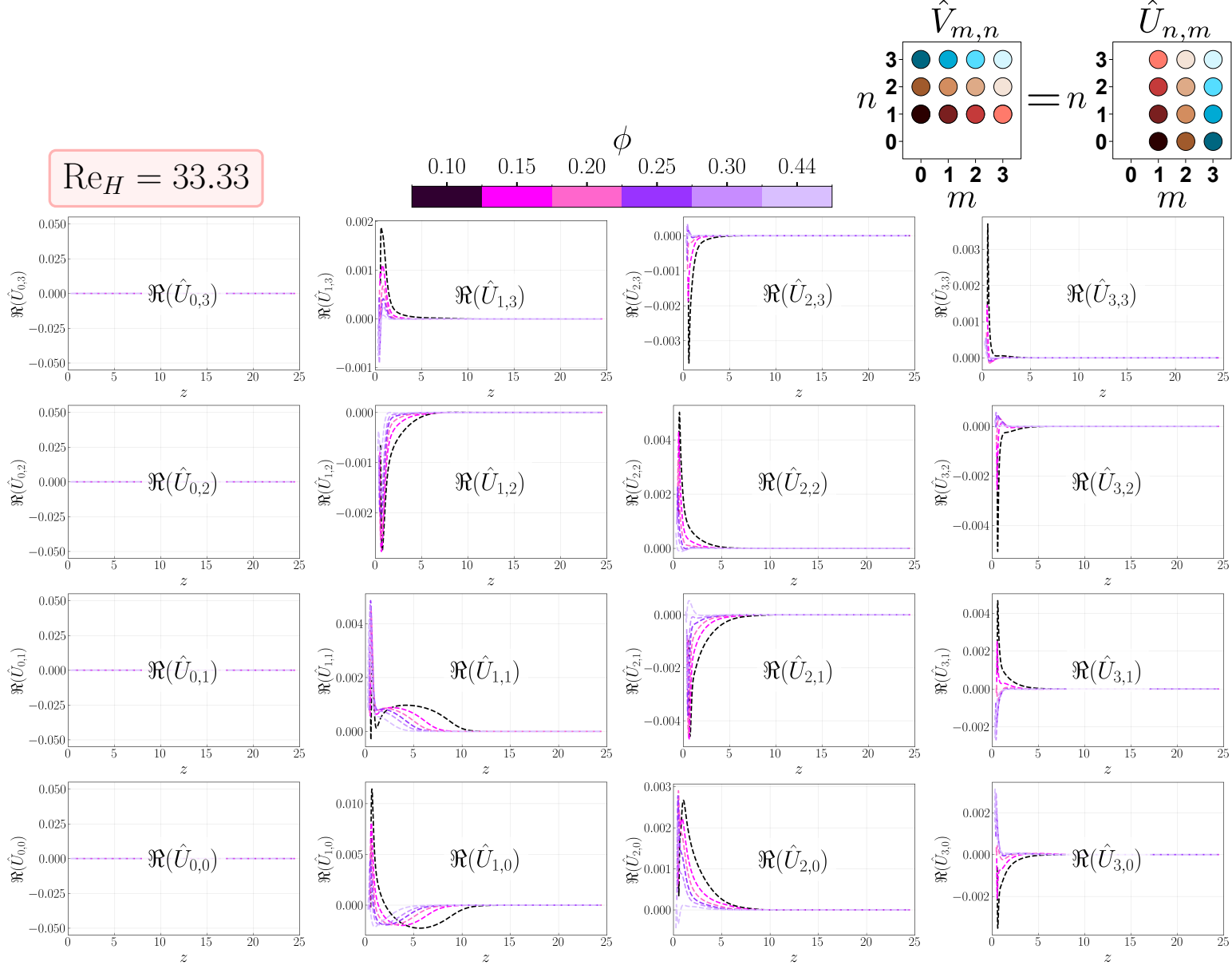


Figure B.55: Streamwise evolution of the real part $\mathcal{R}(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 33.33$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

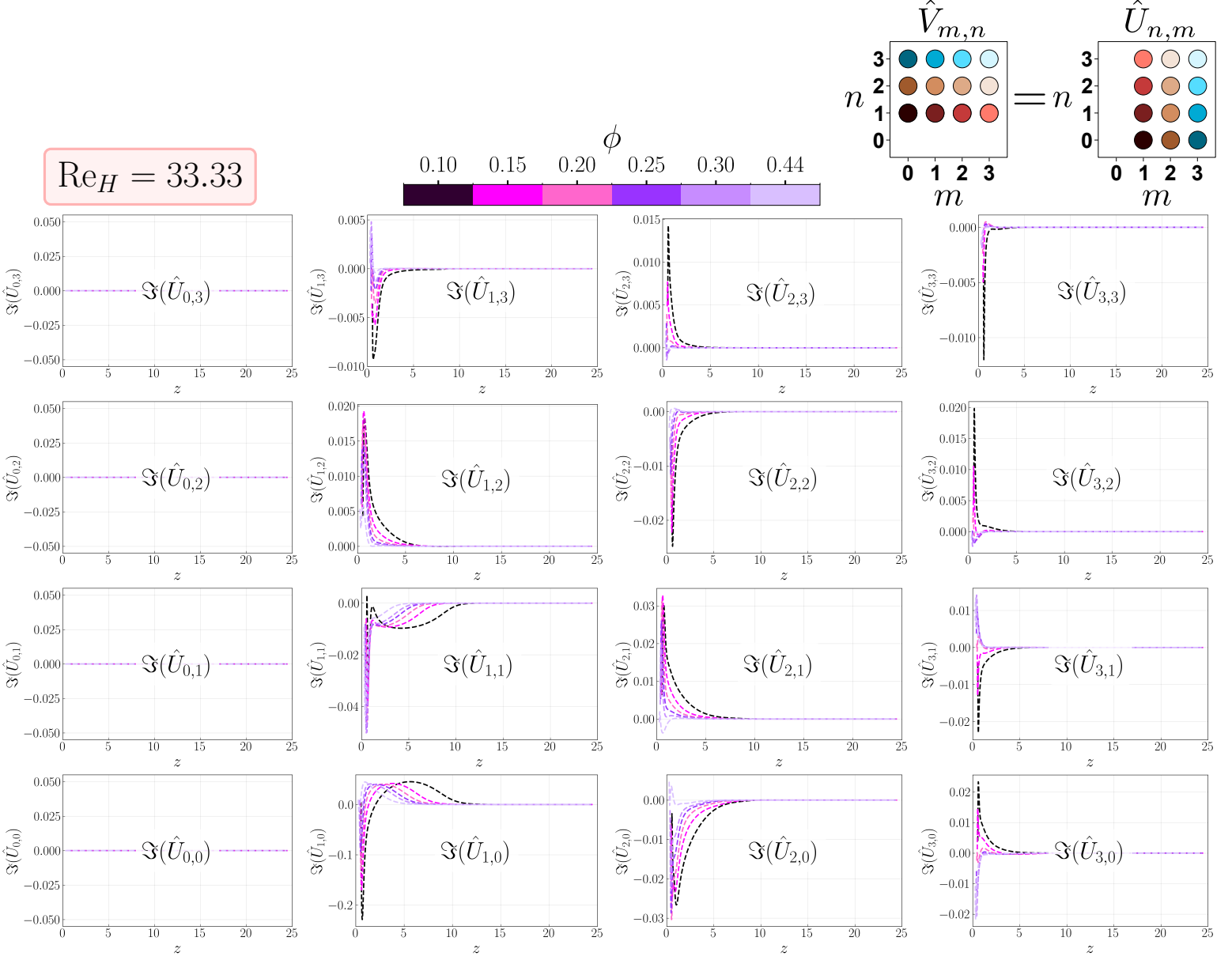


Figure B.56: Streamwise evolution of the imaginary part $\Im(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 33.33$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

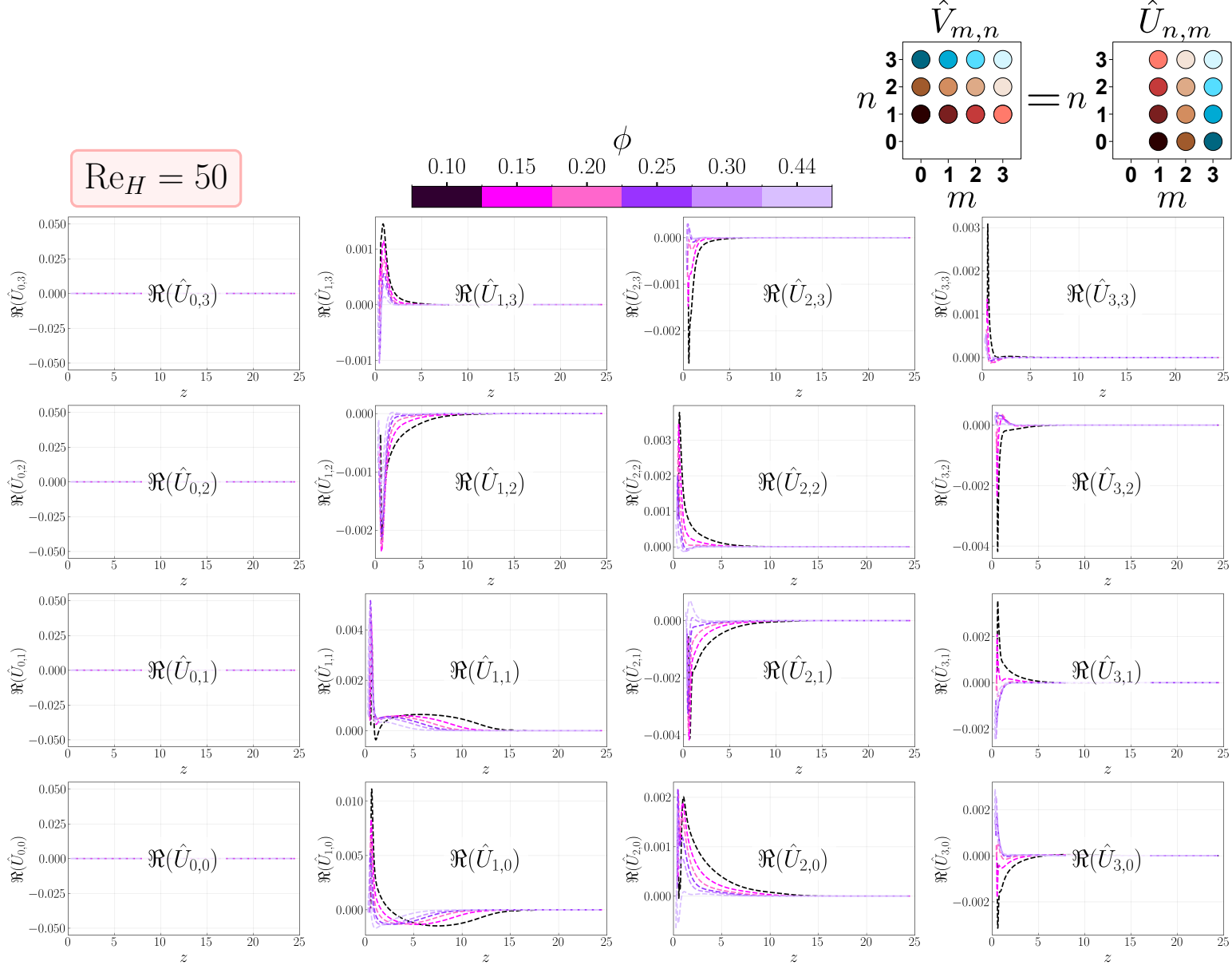


Figure B.57: Streamwise evolution of the real part $\Re(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 50$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

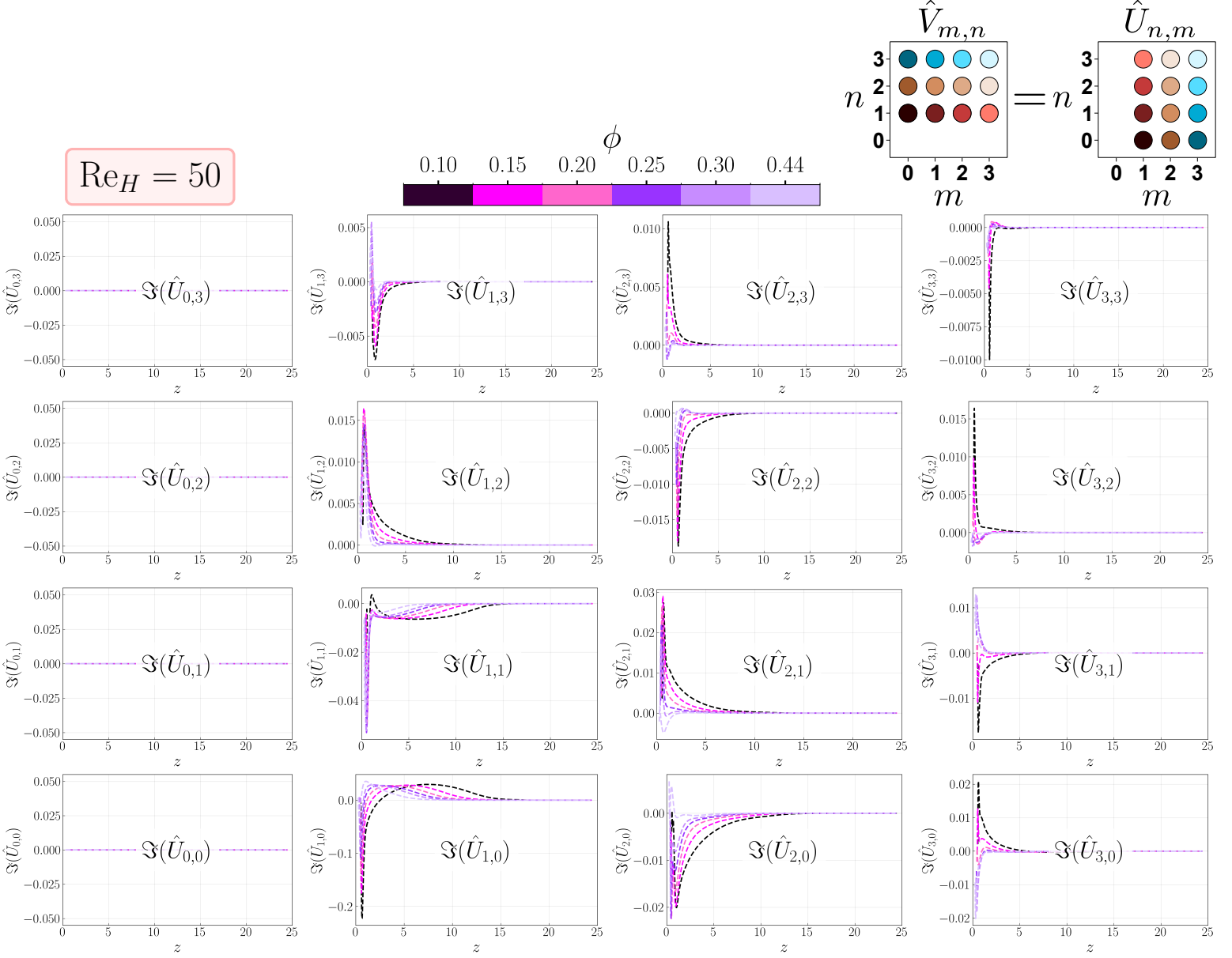


Figure B.58: Streamwise evolution of the imaginary part $\Im(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 50$. The displayed coefficients correspond to the positive Fourier modes $m, n \in \{0, 1, 2, 3\}$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the Fourier modes and highlights the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$.

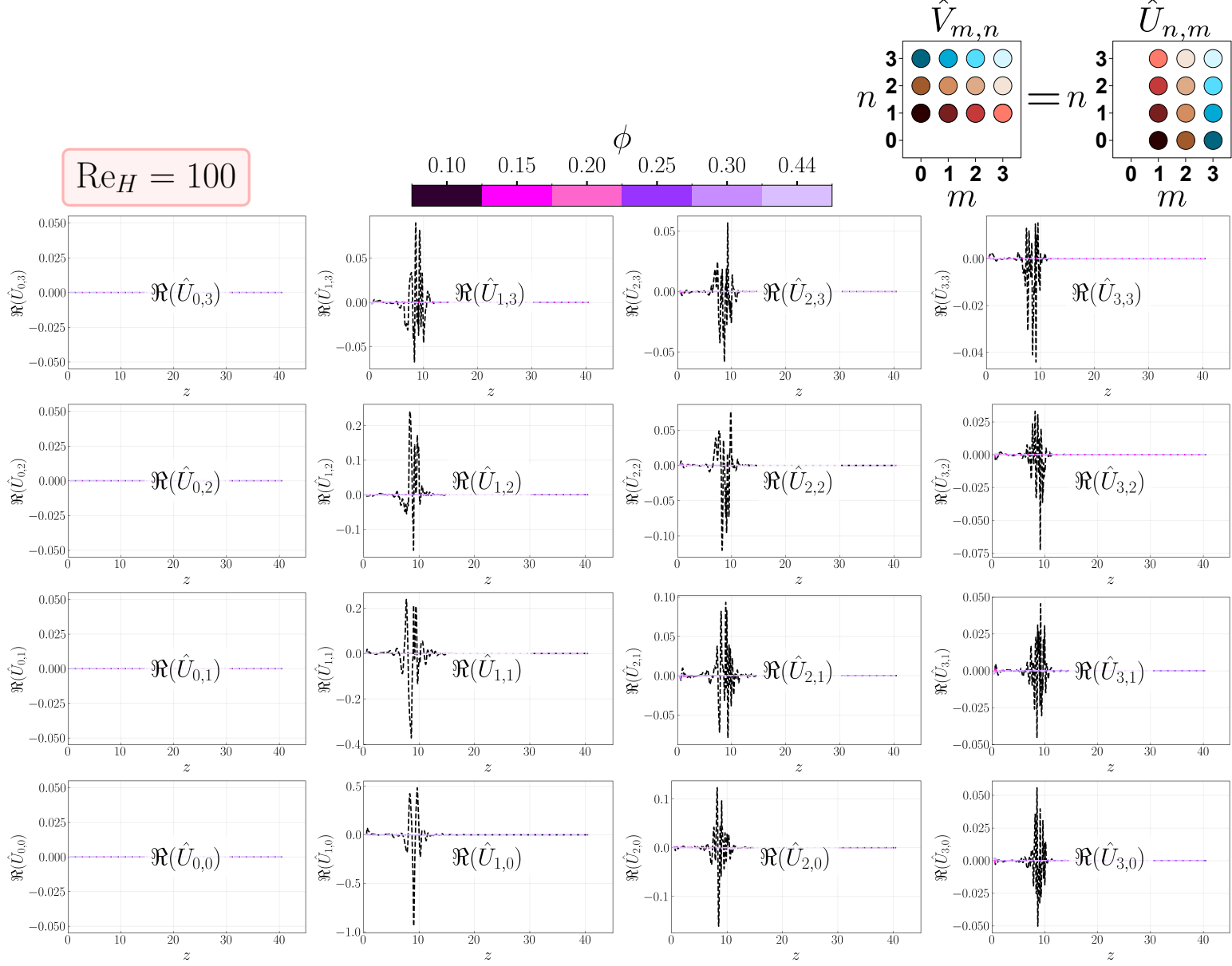


Figure B.59: Streamwise evolution of the real part $\mathcal{R}(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 100$. The displayed coefficients correspond to the positive Fourier modes $m, n \in 0, 1, 2, 3$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the retained Fourier modes and the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$, which holds for the steady-flow cases. Deviations from this symmetry are observed for the unsteady case at $\text{Re}_H = 100$ and $\phi = 0.10$, reflecting the onset of symmetry-breaking dynamics.

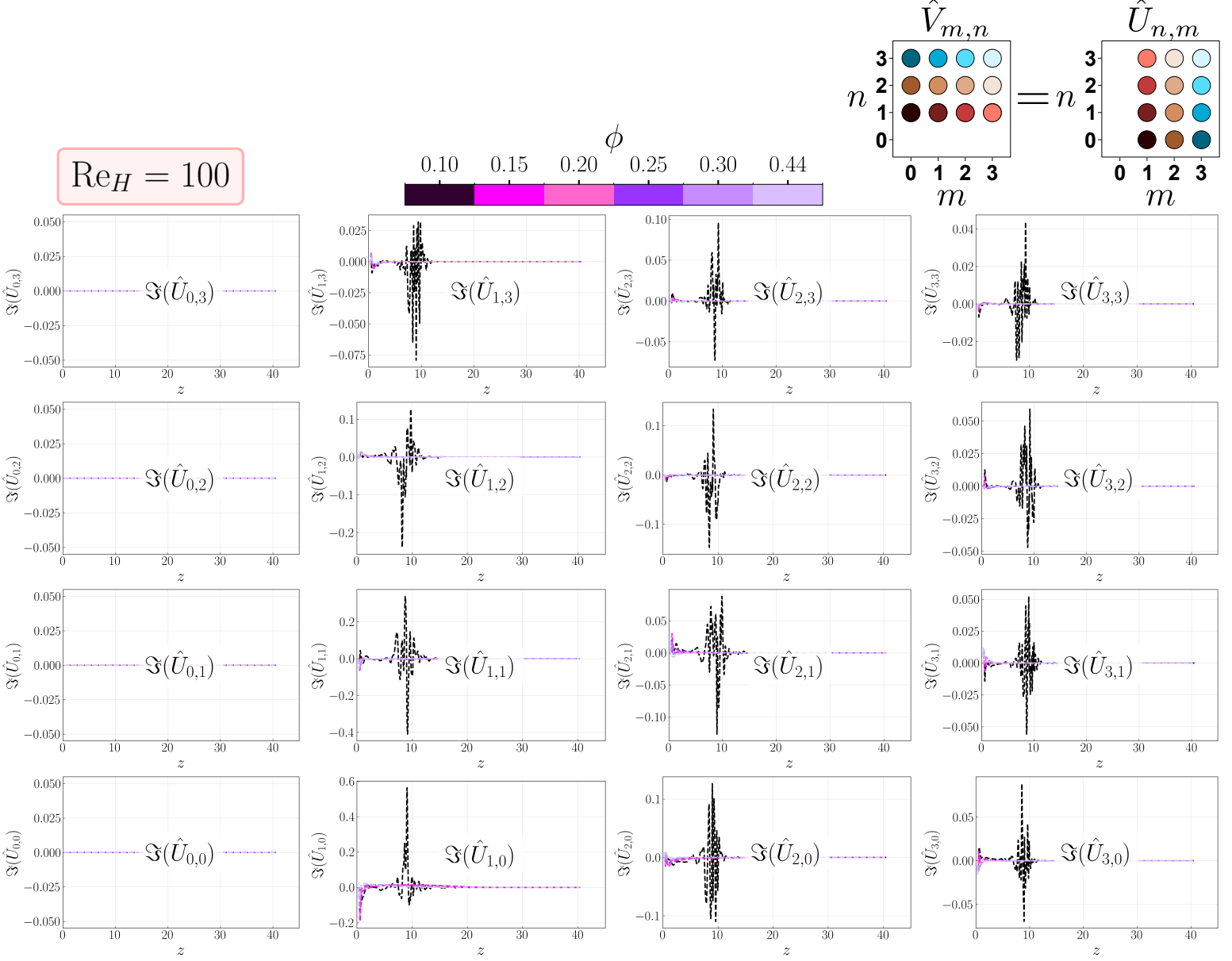


Figure B.60: Streamwise evolution of the imaginary part $\Im(\hat{U}_{m,n})$ of the Fourier coefficients downstream of the SRG at $\text{Re}_H = 100$. The displayed coefficients correspond to the positive Fourier modes $m, n \in 0, 1, 2, 3$, with colors representing the porosity ϕ . The schematic in the upper-right corner illustrates the arrangement of the retained Fourier modes and the symmetry relation $\hat{V}_{m,n} = \hat{U}_{n,m}$, which holds for the steady-flow cases. Deviations from this symmetry are observed for the unsteady case at $\text{Re}_H = 100$ and $\phi = 0.10$, reflecting the onset of symmetry-breaking dynamics.

For SRG flows that preserve the geometrical symmetries, the coefficients satisfy

$$\hat{W}_{m,n} = \hat{W}_{n,m}, \quad \hat{V}_{m,n} = \hat{U}_{n,m}.$$

These relations reduce the number of independent coefficients needed for the spectral representation. They need not hold when the flow breaks the grid symmetry, as observed for the unsteady case at $\phi = 0.10$ and $\text{Re}_H = 100$.

The individual coefficients are combined in the normalized modal energy $E_{m,n}/E_{\text{tot}}$. The following figures show how each mode's share of the total energy evolves downstream and changes with porosity and Reynolds number. They provide a direct view of the spectral compactness of the flow.

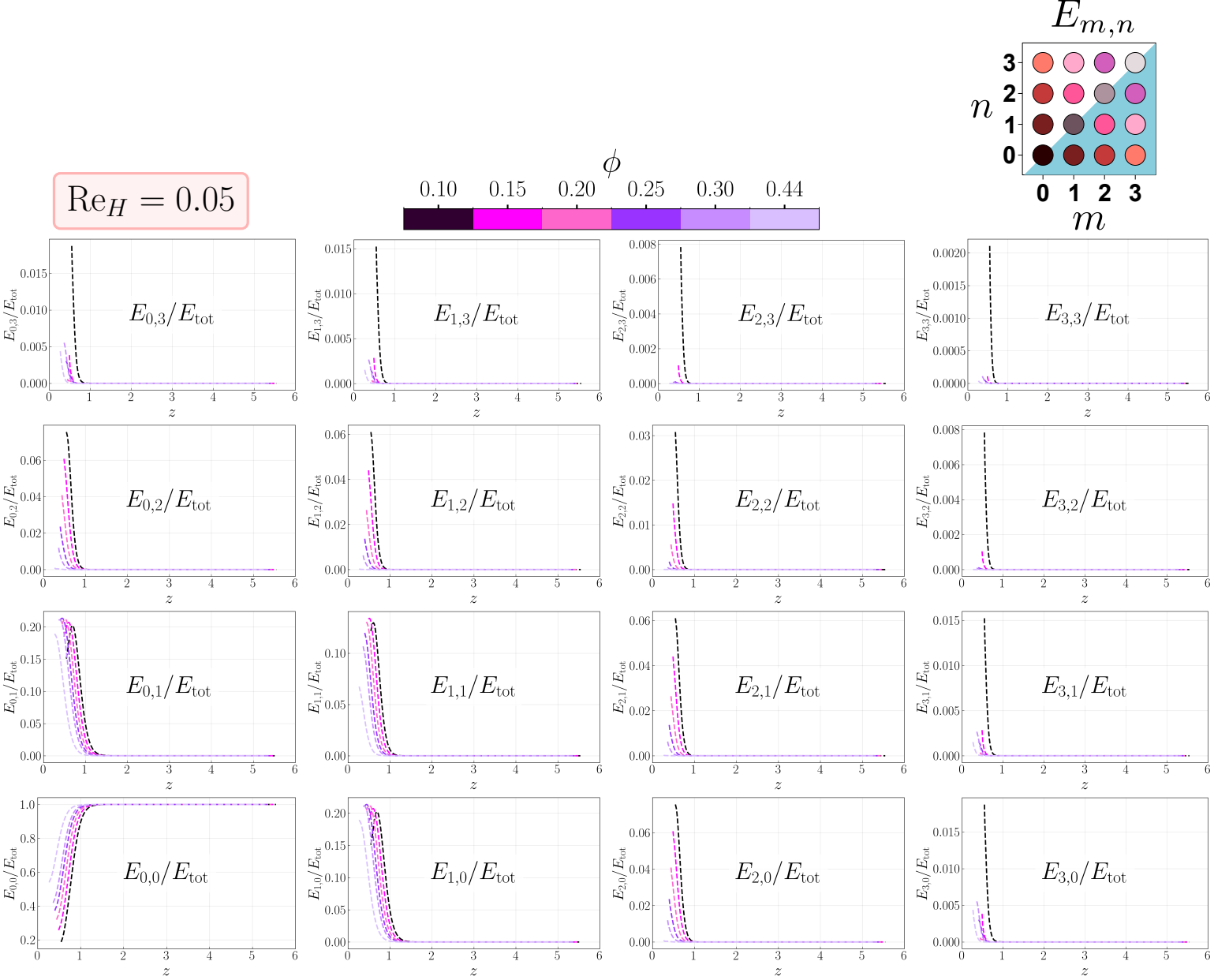


Figure B.61: Streamwise evolution of the normalized modal energy $E_{m,n}/E_{\text{tot}}$ downstream of the SRG at $\text{Re}_H = 0.05$. Each panel shows one Fourier mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property $E_{m,n} = E_{n,m}$.

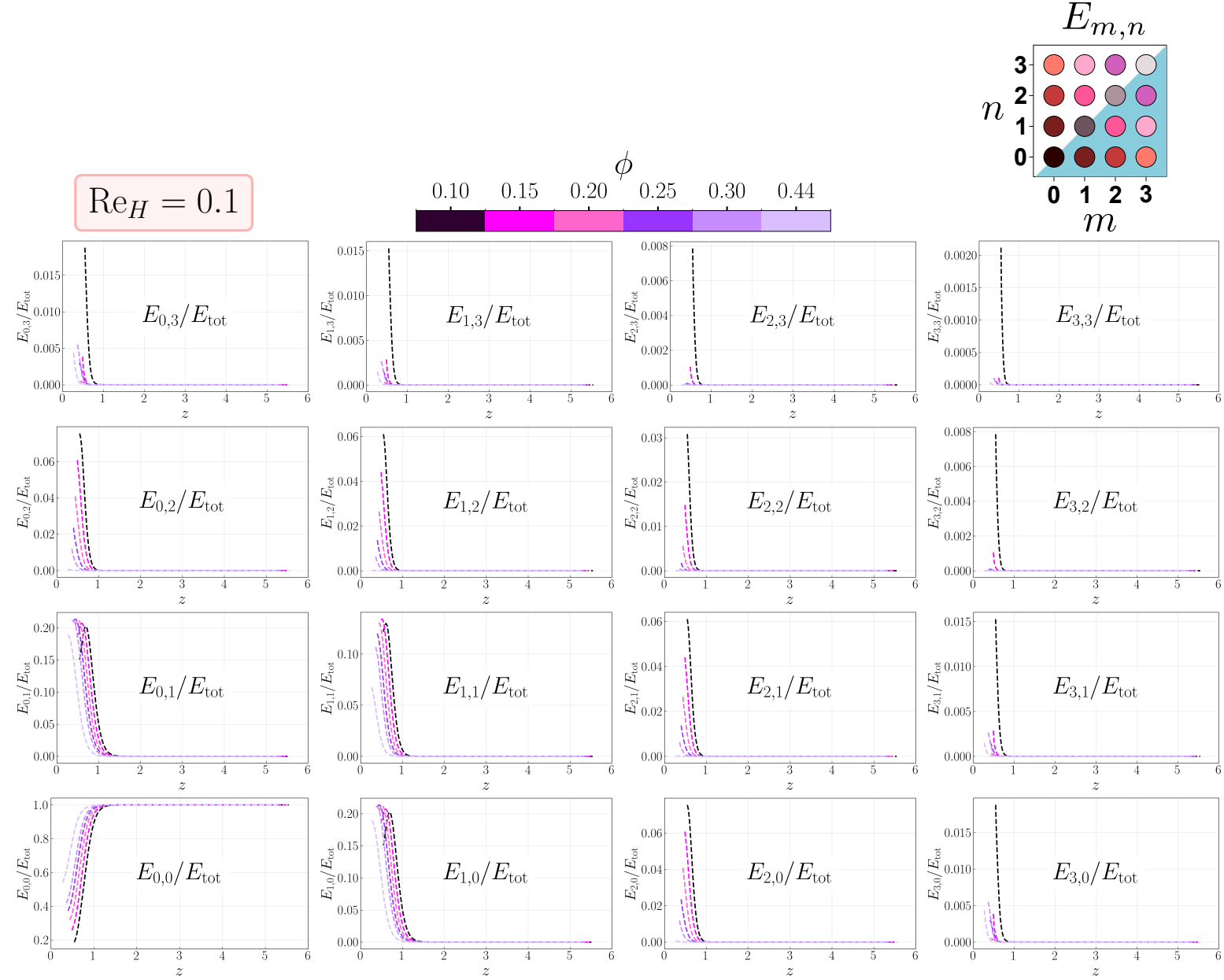


Figure B.62: Streamwise evolution of the normalized modal energy $E_{m,n}/E_{\text{tot}}$ downstream of the SRG at $\text{Re}_H = 0.1$. Each panel shows one Fourier mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property $E_{m,n} = E_{n,m}$.

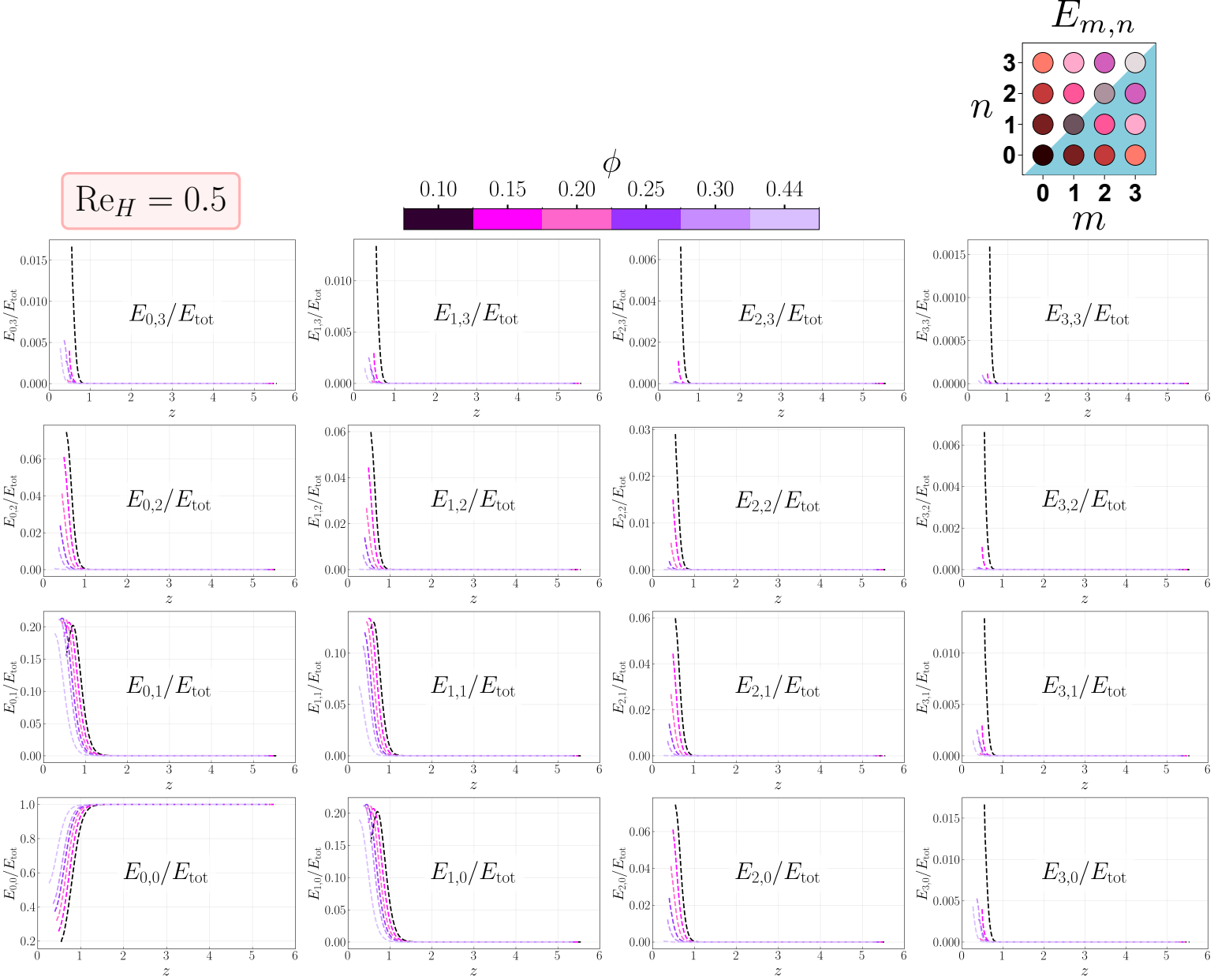


Figure B.63: Streamwise evolution of the normalized modal energy $E_{m,n}/E_{\text{tot}}$ downstream of the SRG at $\text{Re}_H = 0.5$. Each panel shows one Fourier mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property $E_{m,n} = E_{n,m}$.

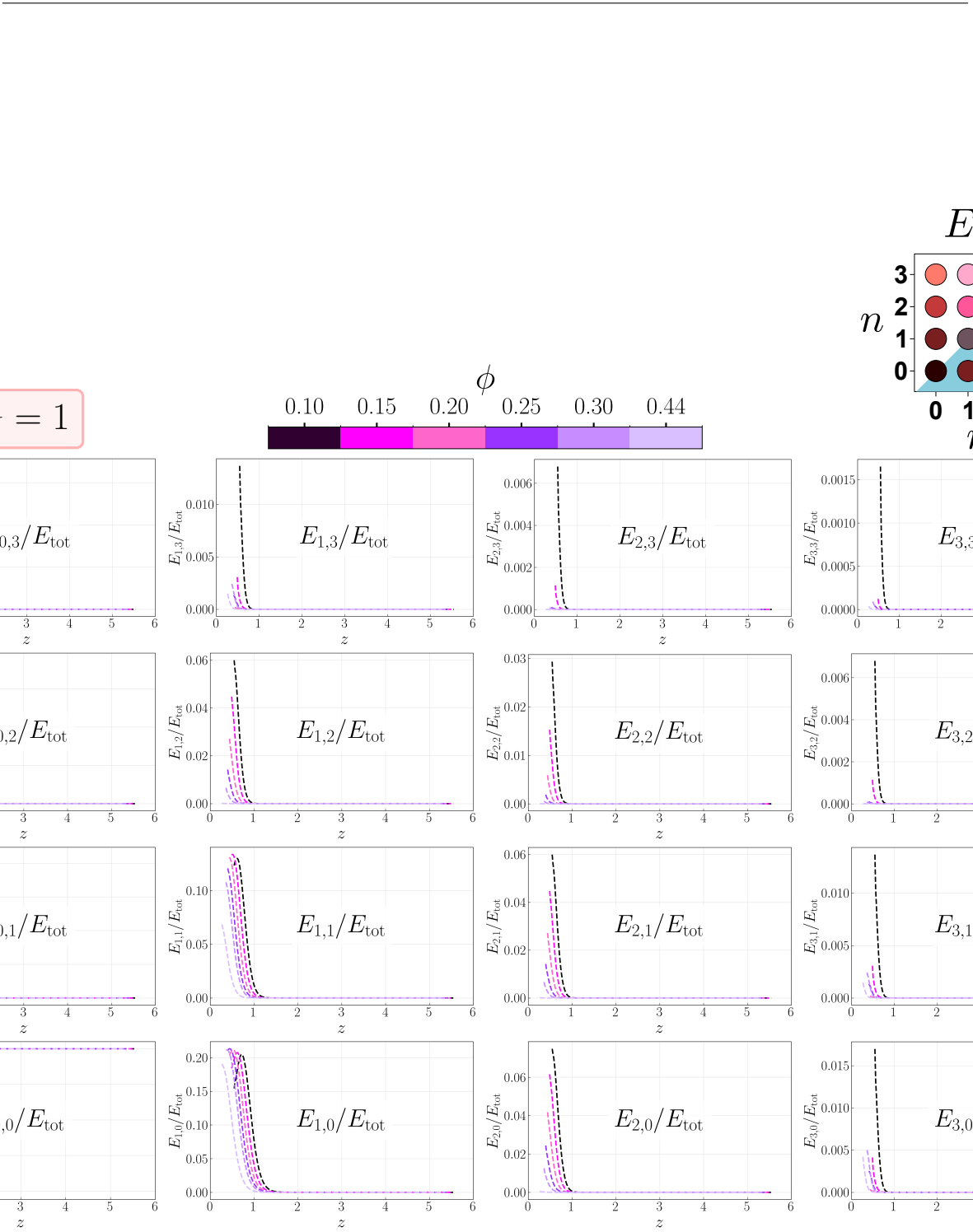


Figure B.64: Streamwise evolution of the normalized modal energy $E_{m,n}/E_{\text{tot}}$ downstream of the SRG at $\text{Re}_H = 1$. Each panel shows one Fourier mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property $E_{m,n} = E_{n,m}$.

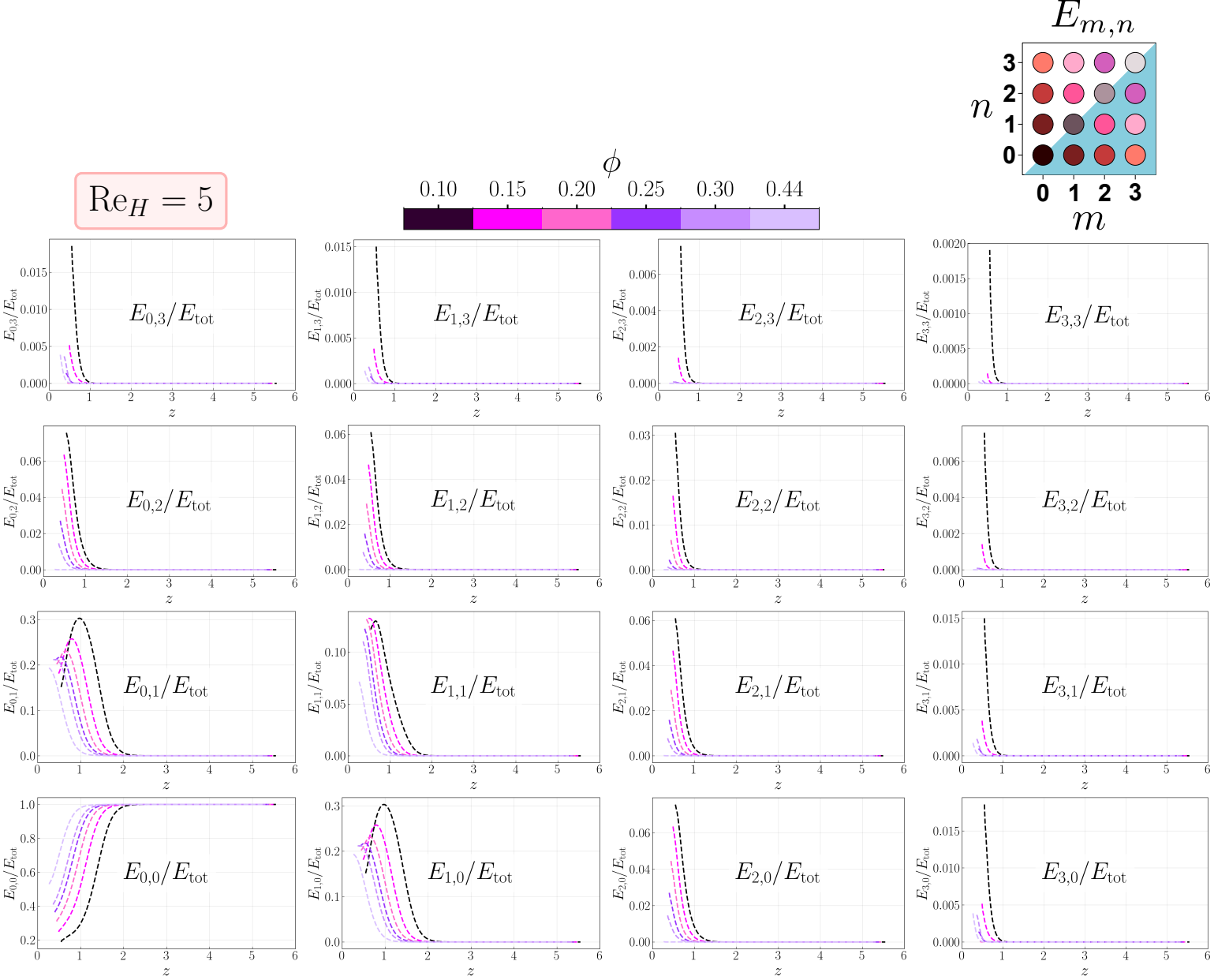


Figure B.65: Streamwise evolution of the normalized modal energy $E_{m,n}/E_{\text{tot}}$ downstream of the SRG at $\text{Re}_H = 5$. Each panel shows one Fourier mode with positive indices $m, n \in \{0, 1, 2, 3\}$; colors indicate the porosity ϕ . The schematic in the upper-right corner shows the arrangement of the retained Fourier modes and highlights the symmetry property $E_{m,n} = E_{n,m}$.